

FEE3411: Control Systems A
Week 1
 Tutorial 1 — Mathematical Toolkit Review

QUESTION SHEET

This hour is a **review**, not new material. Everything here you have met in FEE 222 or FEE 372; the purpose is to get it back into working order, because from Week 2 onwards it is assumed silently in every derivation.

Questions marked [in the hour] are the ones we work through together. Those marked [home-work] you do in your own time — they are not collected, but Week 2 assumes you have done them.

Preparation: Khalil Appendix A (p. 438) or Nise §2.2 (p. 35).

Reading

The toolkit you should be able to use without looking anything up.

Pairs

$\delta(t)$	1
$u(t)$	$1/s$
t	$1/s^2$
$t^n/n!$	$1/s^{n+1}$
e^{-at}	$1/(s+a)$
te^{-at}	$1/(s+a)^2$
$\sin \omega t$	$\omega/(s^2 + \omega^2)$
$\cos \omega t$	$s/(s^2 + \omega^2)$
$e^{-at} \sin \omega t$	$\omega/[(s+a)^2 + \omega^2]$
$e^{-at} \cos \omega t$	$(s+a)/[(s+a)^2 + \omega^2]$

Properties

Linearity	$\mathcal{L}\{af + bg\} = aF + bG$
Differentiation	$\mathcal{L}\{\dot{f}\} = sF(s) - f(0^-)$ $\mathcal{L}\{\ddot{f}\} = s^2F(s) - sf(0^-) - \dot{f}(0^-)$
Integration	$\mathcal{L}\left\{\int_0^t f\right\} = F(s)/s$
Frequency shift	$\mathcal{L}\{e^{-at}f(t)\} = F(s+a)$
Time shift	$\mathcal{L}\{f(t-T)u(t-T)\} = e^{-sT}F(s)$
Initial value	$f(0^+) = \lim_{s \rightarrow \infty} sF(s)$
Final value	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$ <i>only if all poles of $sF(s)$ have negative real parts</i>

Question 1 — Standard test signals

[in the hour]

- Sketch, on the same time axis, the unit impulse $\delta(t)$, unit step $u(t)$, unit ramp $r(t) = tu(t)$ and unit parabola $p(t) = \frac{1}{2}t^2u(t)$. Write down the Laplace transform of each.
- Show that each signal in the list is the time integral of the one before it. Using the integration property only, deduce the transforms $1/s$, $1/s^2$, $1/s^3$ from $\mathcal{L}\{\delta\} = 1$.
- Why does a control engineer test with *these* signals rather than, say, a sine wave or a recorded real input?

Question 2 — Transforms and properties

[in the hour]

- From the definition $F(s) = \int_0^\infty f(t)e^{-st} dt$, derive $\mathcal{L}\{e^{-at}\}$. For which values of s does the integral converge?
- Using the properties in the box, and *without* integrating, find

(i) $\mathcal{L}\{te^{-3t}\}$ (ii) $\mathcal{L}\{e^{-2t} \sin 4t\}$ (iii) $\mathcal{L}\{e^{-(t-2)}u(t-2)\}$

- (c) Find $F(s)$ for $f(t) = 5 - 3e^{-2t} \cos 4t$.
- (d) For $F(s) = \frac{10(s+2)}{s(s^2+4s+13)}$, find $f(0^+)$ and $f(\infty)$ without inverting.
- (e) A student applies the final value theorem to $G(s) = \frac{5}{s^2+9}$ and reports $g(\infty) = 0$. What has gone wrong?

Question 3 — Partial fractions

[in the hour]

Expand each of the following and hence find $f(t)$. In each case check your answer against the initial and final value theorems.

- (a) $F(s) = \frac{s+3}{s(s+1)(s+2)}$ (distinct real poles)
- (b) $F(s) = \frac{s+4}{s(s+2)^2}$ (a repeated pole)
- (c) $F(s) = \frac{10}{s(s^2+2s+5)}$ (a complex pair)

For (c), also read off the undamped natural frequency ω_n and damping ratio ζ of the quadratic factor.

Question 4 — Complex numbers and the s -plane

[in the hour]

This question is the one that pays off most later: the graphical evaluation in part (b) is the whole basis of root-locus sketching in Week 8 and of Bode and Nyquist plots in Weeks 10–12.

Consider

$$G(s) = \frac{4(s+2)}{s(s+4)}.$$

- (a) Mark the poles ($\times$) and zeros ($\circ$) of G on an s -plane diagram.
- (b) Evaluate $G(s)$ at the point $s = -1 + j$ **graphically**: draw the vector from each pole and zero to that point, measure (or compute) its length and angle, and combine them. Confirm your answer algebraically.
- (c) A second-order system has poles at $s = -1 \pm j2$. Write down the form of the corresponding time-domain term. On a sketch of the s -plane, mark ω_n and the angle θ from the negative real axis, and show that $\zeta = \cos \theta$. Evaluate both.
- (d) In one sentence each: what does the *real* part of a pole control, and what does the *imaginary* part control?

Question 5 — Solving a differential equation

[homework]

The position servomechanism of this week's lecture (Figure 5 of the notes), with a particular choice of amplifier gain, obeys

$$\ddot{c} + 4\dot{c} + 8c = 8r(t),$$

where r is the reference shaft angle and c the load shaft angle.

- (a) With the load initially at rest at the origin, $c(0) = \dot{c}(0) = 0$, find $c(t)$ for a unit step reference $r(t) = u(t)$.
- (b) Identify ω_n and ζ . Is the response overdamped, critically damped or underdamped?
- (c) Now set $r(t) = 0$ but start the load displaced: $c(0) = 1$, $\dot{c}(0) = 0$. Find $c(t)$.
- (d) Comment on what is the same and what is different between (a) and (c), and say which part of the answer is a property of the *system* rather than of the input.

Question 6 — Time delay, and a warning about $s \rightarrow \infty$ [homework]

- (a) A sensor introduces a pure time delay of T seconds: its output is the input, unchanged in shape, but arriving T seconds late. Show from the time-shift property that its transfer function is e^{-sT} .
- (b) A step of height 5 is applied at $t = 0$ to such a sensor with $T = 0.4$ s. Write down the transform of the output, and sketch the output against time.
- (c) Explain why e^{-sT} is *not* a ratio of polynomials in s , and why that is inconvenient. (You are not expected to solve this problem yet — it is dealt with in Weeks 3 and 10.)
- (d) Apply the initial value theorem to $F(s) = \frac{s+1}{s+2}$. Then find $f(t)$ properly and explain the discrepancy.